

LAB 2 [Group # → 1 to 8]

[Platform → HACKERRANK]

Introduction To Programming

(21-01-22)

QUESTIONS

Q1 → TRIBONACCI SEQUENCE

Given an integer value **N**, find the **Nth** term of a *Tribonacci sequence*.

Tribonacci sequence is a sequence in which each term is the sum of the previous 3 terms i.e. 0, 1, 1, 2, 4, 7, 13, 24, 44, 81...

The first, the second and the third term in the Tribonacci sequence are 0, 1 and 1 respectively.

$$T[1] = 0, T[2] = 1, T[3] = 1$$

$$T[4] = T[1] + T[2] + T[3] = 0 + 1 + 1 = 2 \Rightarrow 4\text{th term}$$

$$T[5] = T[2] + T[3] + T[4] = 1 + 1 + 2 = 4 \Rightarrow 5\text{th term}$$

Input Format

The first and only input line contains **N**

Constraints

$$1 \leq N \leq 100$$

Output Format

Output is an integer denoting the **Nth** Tribonacci Number.

Sample Input 0

1

Sample Output 0

0

Explanation 0

As the first term of Tribonacci sequence is 0, the output must be 0

Sample Input 1

7

Sample Output 1

13

Explanation 1

The 7th term in Tribonacci sequence is 13

Q2 → SMALLEST 2

You met a Genie who grants wishes. But before granting wishes, Genie wants to know whether you are worthy enough or not.

Prove yourself by solving the following problem :

Given ' n ' integers, find the smallest two integers.

Input Format

First line contains an integer ' n '.

Each of the n next lines contains a single integer x .

Constraints

$2 \leq n \leq 1000$

$1 \leq x \leq 100000$

Output Format

Print the **SMALLEST** 2 numbers out of the given numbers.

First print the smallest one and then, print the next smallest one, followed by a space.

Sample Input 0

5
5
4
3
2
1

Sample Output 0

1 2

Explanation 0

The first smallest number is :1

The second smallest number is :2

Sample Input 1

```
6
6
3
3
4
5
10
```

Sample Output 1

```
3 3
```

Explanation 1

Both the first smallest and second smallest number, is : 3

Q3 → FRESHERS PARTY

Your college seniors are organizing a Fresher's Party tomorrow. You want to go to that party with your friends. But your mom is in a bad mood today. She will not allow you to the party if she is in a bad mood. Solve this question to make your mom happy :

You are given four numbers ***a***, ***b***, ***c*** and ***d***. Print the product of these numbers. If there are duplicates present, consider them only once.

See below explanation for more details.

Input Format

The first line contains an integer ***a***.

The second line contains an integer ***b***.

The third line contains an integer ***c***.

The forth line contains an integer ***d***.

Constraints

$0 \leq a, b, c, d \leq 10000$

Output Format

Print the product

Sample Input 0

```
2
3
4
5
```

Sample Output 0

```
120
```

Explanation 0

Product = $2 * 3 * 4 * 5 = 120$, all numbers are distinct

Sample Input 1

```
3
4
3
5
```

Sample Output 1

```
60
```

Explanation 1

Product = $3 * 4 * 5 = 60$, Note number 3 is a duplicate and hence consider it only once.

Q4 → BABYLONIAN ALGORITHM

The **Babylonian** algorithm is one of the ancient methods to compute the *square root* of a natural number, **N** within some permissible error. It works as follows:

1. An initial guess, say ***g_current*** is made for the square root.
2. The next guess, say ***g_next*** is computed as the average of ***g_current*** and $N/g_current$, i.e, ***g_next*** = $(g_current + (N/g_current))/2$.
3. Step 2 is repeated till the difference between successive estimates is less than or equal to 0.0001, i.e, ***g_next*** - ***g_current*** ≤ 0.0001

Input Format

The first line contains an input for ***N***.

The second line contains an input for initial estimate, ***g_current***.

Constraints

$0 \leq N \leq 1000$

Output Format

The output contains the estimated square root of **N**, upto 2 *decimal places*.

[**HINT**: You can use string formatting for this]

Sample Input 0

```
17
3.0
```

Sample Output 0

```
4.13
```

Explanation 0

Square root estimate from the babylonian method for 17 has been found to be 4.13.

Sample Input 1

```
20
4
```

Sample Output 1

```
4.47
```

Explanation 1

The square root estimate for 20 using the mentioned method has been found to be 4.47

Q5 → LEAP YEAR

Ram wants to check whether the year is a leap year or not. So you have to write a program to check whether the year is a leap year or not.

Input Format

The only line of input consists of a year **N**

Constraints

$1 \leq N \leq 10^6$

Output Format

YES if N is a leap year otherwise **NO**

Sample Input 0

```
2016
```

Sample Output 0

```
YES
```

Sample Input 1

1900

Sample Output 1

NO

Q6 → FIND DIVISORS[NOT GRADED]

For a given number 'n', find all its divisors. A divisor of a number **n** is a positive integer that divides it without leaving a remainder.

Input Format

The first line contains a positive integer **n**.

Constraints

$1 \leq n \leq 1000$

Output Format

In subsequent lines, print divisors of n in increasing order.

Sample Input 0

12

Sample Output 0

1
2
3
4
6
12

Explanation 0

The factors of 12 are 1, 2, 3, 4, 6, 12. These factors have been printed out in each of the lines respectively.

Q7 → WHAT'S YOUR BONUS[NOT GRADED]

A company decided to give bonus to its employees according to the following criteria:

Time period of Service	Bonus
------------------------	-------

More than 8 years	15%
≥ 4 and ≤ 8 years	10%
Less than 4 years	5%

The bonus percentage is calculated on the salary of the employee.

The problem is to find the *bonus* amount for an employee given his salary and years of service in the company.

Input Format

First line contains an integer, *years of service*

Second line contains an integer, *salary*

Constraints

$20000 \leq \text{salary} \leq 500000$

$0 \leq \text{years of service} \leq 60$

Output Format

An integer.

Contains the bonus amount rounded to the greatest integer value smaller than the actual amount.(eg. if the actual bonus is 1222.57, then output should be 1222)

Sample Input 0

```
9
40000
```

Sample Output 0

```
6000
```

Explanation 0

Since *years of service* is greater than 8, the bonus would be 15 % of the salary. 15% of 40000 is 6000, which is the answer.